The empirical recurrence for OEIS A223972: recovering a conjecture that is not written down

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Abstract

OEIS A223972 records "Empirical recurrence of order 43", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 273$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 43, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 43 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A223972 is "Number of nX5 0..2 arrays with rows and antidiagonals unimodal.". It has offset 1 and begins

$$86, 7396, 440628, 22935921, 1140963027, 55803232969, 2708281019793, 131014406127439, \ldots$$

The entry states:

> Empirical recurrence of order 43 (see link above)

As of the "Last modified" line on the live entry (Oct 03 2025 23:05:45, revision 10) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2752$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 273$ of the 2752, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 273$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 546$ exact terms of the model returns order 43 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{43} c_i\, a(n-i),$$

c_1	86	c_{12}	-1726647214238	c_{23}	-365986078954188943	c_{34}	$-9118249590798422348 8$
c_2	-2331	c_{13}	2787245634530	c_{24}	-895781156538954197	c_{35}	-20306930947685369856
c_3	30139	c_{14}	20826056854339	c_{25}	611652785159644541 3	c_{36}	67366072582569768960
c_4	-326058	c_{15}	-136388205061356	c_{26}	$-1499829760487255731 6$	c_{37}	-38944600927931068416
c_5	2831199	c_{16}	541254283458013	c_{27}	18117365743200996120	c_{38}	21225654594186117 12
c_6	6915227	c_{17}	-739403436297281	c_{28}	5916352808535583024	c_{39}	8364032046189379584
c_7	-351300717	c_{18}	-5246910125344719	c_{29}	-65279569581530331960	c_{40}	-4635856329375744000
c_8	2106768137	c_{19}	27794848207567973	c_{30}	108164135596772303152	c_{41}	1160985231605366784
c_9	-2659493617	c_{20}	-56207720490279751	c_{31}	-55951774403026410432	c_{42}	-144813768541470720
c_{10}	-10549211272	c_{21}	30120520616637344	c_{32}	-72807082570180422144	c_{43}	7264970735616000
c_{11}	205885332238	c_{22}	165768413686317367	c_{33}	148015577792292691072		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 43, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $43+4$ terms removed returns order 43 as well, so $\deg q_{\min} = 43 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $43+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 43 that this sequence satisfies — and that family turns out to have exactly one member.

References

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